



Savitribai Phule Pune University		
Fourth Year of Computer Engineering (2019 Course)		
410250: High Performance Computing		
Teaching Scheme: TH: 4 Hours/Week	Credit 3	Examination Scheme:100 Mid-Semester (TH) : 30 End- Sem (TH): 70
Prerequisites Courses: -Microprocessor (210254), Principles of Programming Languages(210255), Computer Networks and Security(310244)		
Companion Course: Laboratory Practice V(410254)		
Course Objectives: - To understand different parallel programming models - To analyze the performance and modeling of parallel programs - To illustrate the various techniques to parallelize the algorithm - To implement parallel communication operations. - To discriminate CUDA Architecture and its components. - To Understand Scope of Parallel Computing and its search algorithms.		
Course Outcomes: CO1: Understand various Parallel Paradigm CO2: Design and Develop an efficient parallel algorithm to solve given problem CO3: Illustrate data communication operations on various parallel architecture CO4: Analyze and measure performance of modern parallel computing systems CO5: Apply CUDA architecture for parallel programming CO6: Analyze the performance of HPC applications		
Course Contents		
Unit I	Introduction to Parallel Computing	09 Hours
Introduction to Parallel Computing: Motivating Parallelism, Modern Processor: Stored-program computer architecture, General-purpose Cache-based Microprocessor architecture. Parallel Programming Platforms: Implicit Parallelism, Dichotomy of Parallel Computing Platforms, Physical Organization of Parallel Platforms, Communication Costs in Parallel Machines. Levels of parallelism, Models: SIMD, MIMD, SIMT, SPMD, Data Flow Models, Demand-driven Computation, Architectures: N-wide superscalar architectures, multi-core, multi-threaded.		
#Exemplar/Case Studies	Case study: Multi-core System	
*Mapping of Course Outcomes for Unit I	CO1	
Unit II	Parallel Algorithm Design	09 Hours
Global System for Mobile Communications (GSM) architecture , Mobile Station, Base Station System, Switching subsystem, Security, Data Services, HSCSD, GPRS - GPRS system and protocol architecture 2.3 UTRAN, UMTS core network; Improvements on Core Network, 802.11 Architecture 802.11a, 802.11b standard		
#Exemplar/Case Studies	IPoC: A New Core Networking Protocol for 5G Networks.	

*Mapping of Course Outcomes for Unit II	CO2	
Unit III	Parallel Communication	09 Hours
Basic Communication: One-to-All Broadcast, All-to-One Reduction, All-to-All Broadcast and Reduction, All-Reduce and Prefix-Sum Operations, Collective Communication using MPI: Scatter, Gather, Broadcast, Blocking and non blocking MPI, All-to-All Personalized Communication, Circular Shift, Improving the speed of some communication operations.		
#Exemplar/Case Studies	Case study: Monte-Carlo Pi computing using MPI	
*Mapping of Course Outcomes for UnitIII	CO3	
Unit IV	Analytical Modeling of Parallel Programs	09 Hours
Sources of Overhead in Parallel Programs, Performance Measures and Analysis: Amdahl's and Gustafson's Laws, Speedup Factor and Efficiency, Cost and Utilization, Execution Rate and Redundancy, The Effect of Granularity on Performance, Scalability of Parallel Systems, Minimum Execution Time and Minimum Cost, Optimal Execution Time, Asymptotic Analysis of Parallel Programs. Matrix Computation: Matrix-Vector Multiplication, Matrix-Matrix Multiplication.		
#Exemplar/Case Studies	Case study: The DAG Model of parallel computation	
*Mapping of Course Outcomes for UnitIV	CO4	
Unit V	CUDA Architecture	09 Hours
Introduction to GPU: Introduction to GPU Architecture overview, Introduction to CUDA C- CUDA programming model, write and launch a CUDA kernel, Handling Errors, CUDA memory model, Manage communication and synchronization, Parallel programming in CUDA- C.		
#Exemplar/Case Studies	Case study: GPU applications using SYCL and CUDA on NVIDIA	
*Mapping of Course Outcomes for Unit V	CO5	
Unit VI	High Performance Computing Applications	09 Hours
Scope of Parallel Computing, Parallel Search Algorithms: Depth First Search(DFS), Breadth First Search(BFS), Parallel Sorting: Bubble and Merge, Distributed Computing: Document classification, Frameworks – Kuberbets, GPU Applications, Parallel Computing for AI/ ML		
#Exemplar/Case Studies	Case study: Disaster detection and management/ Smart Mobility/Urban planning	
*Mapping of Course Outcomes for Unit VI	CO6	
Learning Resources		

Text Books:

1. Ananth Grama, Anshul Gupta, George Karypis, and Vipin Kumar, "Introduction to Parallel Computing", 2nd edition, Addison-Wesley, 2003, ISBN: 0-201-64865-2
2. Seyed H. Roosta, "Parallel Processing and Parallel Algorithms Theory and Computation", Springer-Verlag 2000, ISBN 978-1-4612-7048-5 ISBN 978-1-4612-1220-1
3. John Cheng, Max Grossman, and Ty McKercher, "Professional CUDA C Programming", John Wiley & Sons, Inc., ISBN: 978-1-118-73932-7

Reference Books :

1. Kai Hwang, "Scalable Parallel Computing", McGraw Hill 1998.
2. George S. Almasi and Alan Gottlieb, "Highly Parallel Computing", The Benjamin and Cummings Pub. Co., Inc
3. Jason Sanders, Edward Kandrot, "CUDA by Example", Addison-Wesley, ISBN-13: 978-0-13-138768-3
4. Pacheco, Peter S., "An Introduction to Parallel Programming", Morgan Kaufmann Publishers ISBN 978-0-12-374260-5
5. Rieffel WH, EG, Polak, "Quantum Computing: A gentle introduction", MIT Press, 2011, ISBN 978-0-262-01506-6
6. Ajay D. Kshemkalyani, Mukesh Singhal, "Distributed Computing: Principles, Algorithms, and Systems", Cambridge March 2011, ISBN: 9780521189842

e Books :

1. http://prdrklaina.weebly.com/uploads/5/7/7/3/5773421/introduction_to_high_performance_computing_for_scientists_and_engineers.pdf
2. https://www.vssut.ac.in/lecture_notes/lecture1428643084.pdf

NPTEL/YouTube video lecture link

- <https://nptel.ac.in/courses/106108055>
- <https://www.digimat.in/nptel/courses/video/106104120/L01.html>

@The CO-PO Mapping Matrix

CO/ PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1	-	-	-	-	-	-	-	-	-	-
CO2	2	1	-	-	-	-	-	-	-	-	-	-
CO3	2	1	-	-	-	-	-	-	-	-	-	-
CO4	1	2	-	2	-	-	-	-	-	-	-	-
CO5	1	2	-	2	-	-	-	-	-	-	-	1
CO6	2	2	-	2	-	-	-	-	-	-	-	1